

LED LIGHTING FOR COMMERCIAL APPLICATIONS

Today, commercial LEDs emit 110-120 lumens/watt and recently, the Bilkent University in Turkey has claimed to have developed an LED with an output of 300 lumens/watts. As the lumens/watt increases, the number of LEDs to be used to replace conventional lighting is coming down

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Light-emitting diode (LED) is gaining popularity as the lighting technology of choice for many commercial and industrial applications. This is because it is much safer, cheaper and energy-efficient than other lighting sources. LED lighting fixtures usually have higher luminance and yet they consume lesser energy, thus inspiring many commercial and industrial establishments to convert into LED lighting. Companies that currently use this technology report that they consume approximately 48 per cent lesser energy than with conventional lighting technology. LED lighting is visually appealing as well.

LED technology is therefore being constantly upgraded to meet the requirements of newer applications. "For commercial purposes, the most effective LED lighting would be MR16, MR11 and downlights of various wattages. These are used in showrooms, shopping malls, offices, factories, etc," says Ravi A.V., general manager, LED division, Nichia Chemical (I).

What's new in the market

Different types of LED lighting products are available in the market, ranging from retrofitted bulbs, tubelights, and flexible and rigid strip lights to a number of downlights and spotlights.

An 'LED product' generally means a product with an upgraded version of LED technology and some specification changes in its components to make it more energy-efficient. The pace of tech-



(Clockwise from top-left) LED projects by Candela Lighting, Binay Opto, Amptek LED Lighting and Sujana Energy